

Python Web Scraping and Analysis Project



Main goal

- ▶ The main goal of your analysis is **to enhance search engine optimization (SEO) strategies by identifying key opportunities for improving website visibility, optimizing on-page elements, and increasing organic traffic, ultimately leading to higher conversion rates and improved overall performance in search rankings.**

Table of contents

1. [Obtaining the SerpAPI Key](#)
2. [Obtaining the NomicAPI Key](#)
3. [Initial Configuration](#)
4. [Analyses in the project](#)
 - 4.1. [Meta Description Length Analysis](#)
 - 4.2. [Keyword in Meta Description Analysis](#)
 - 4.3. [Keyword in Title Analysis](#)
 - 4.4. [Title Length Analysis](#)
 - 4.5. [Duplicate Title Analysis](#)
 - 4.6. [URL Length Analysis](#)
 - 4.7. [Parameteres in URL Analysis+](#)
5. [Project summary](#)
6. [Business conclusions](#)



To get started, you need two things:

1) the SerpAPI key



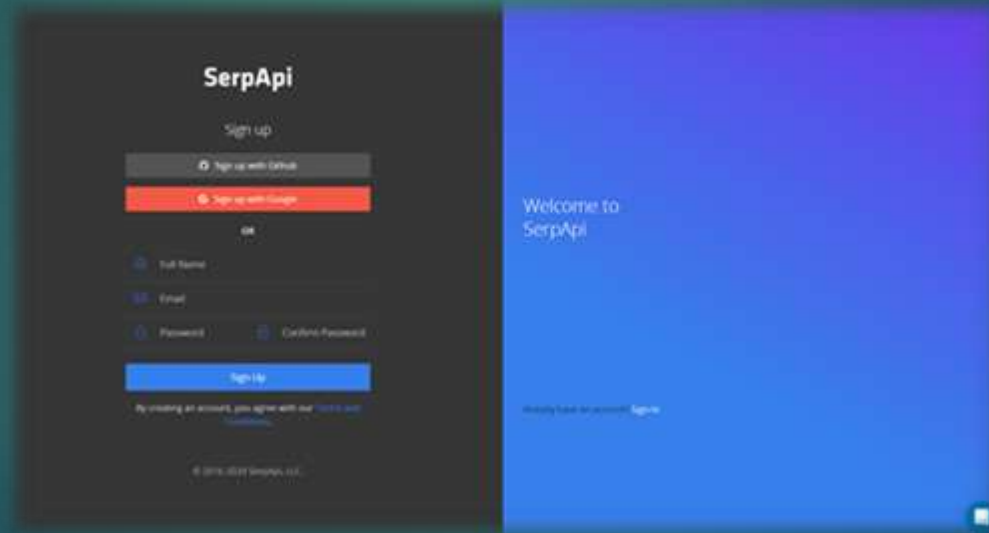
2) the NomicAPI key

NOMIC



Obtaining the SerpAPI Key

1. Go to the website https://serpapi.com/users/sign_up if you're using this service for the first time, please sign up with Google





Obtaining the SerpAPI Key

2. You will receive a registration confirmation email, and you need to click the "Confirm your email" button.

You can confirm your email through the link below:

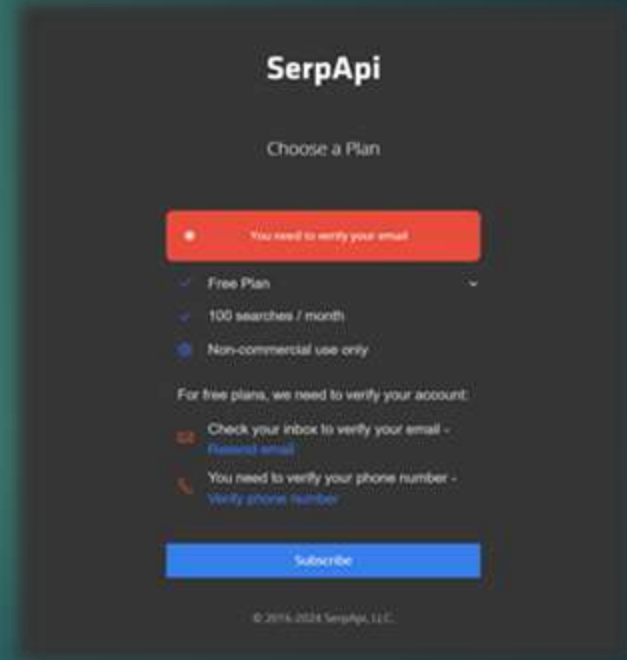
Confirm your email

Thank you!
SerpApi Team



Obtaining the SerpAPI Key

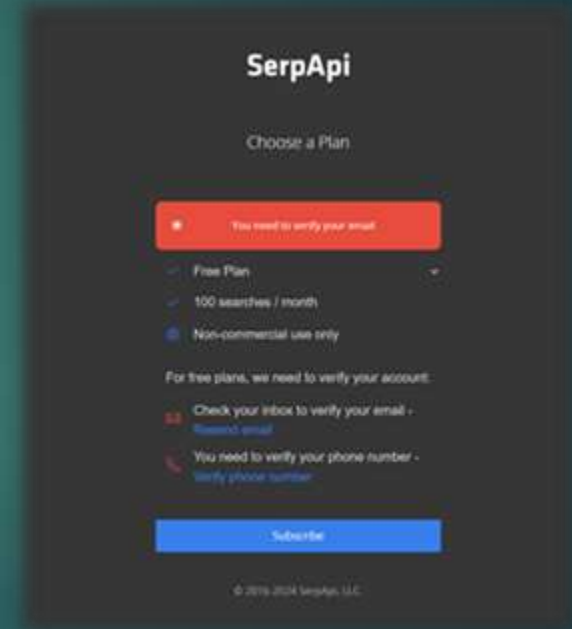
3. You will be redirected to <https://serpapi.com/users/welcome> , and you will need to choose a service plan. For our needs, the free plan will suffice. Select it and click the "Subscribe" button.





Obtaining the SerpAPI Key

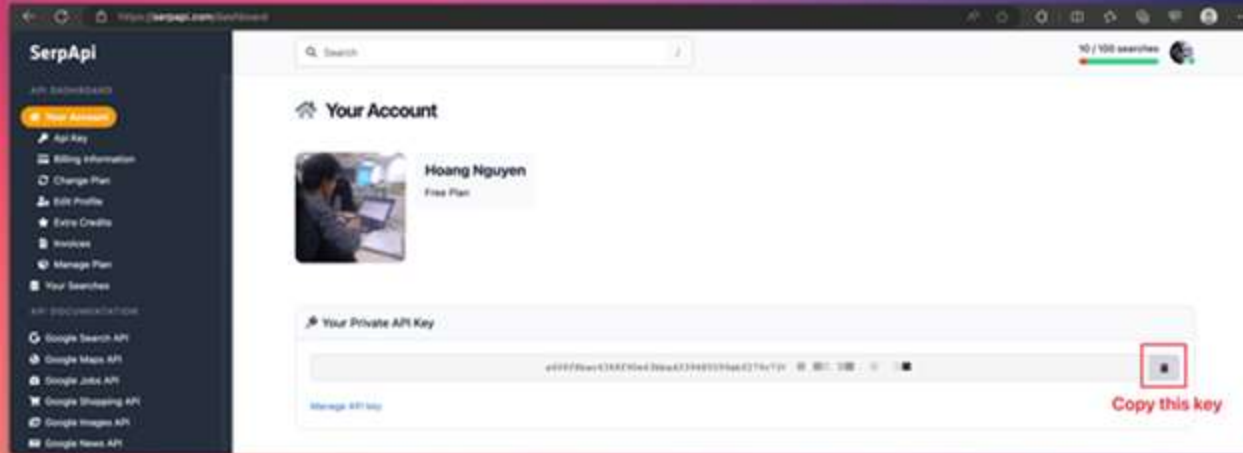
4. Next, you need to verify your phone number. Click "Verify phone number," enter your phone number, and click the "Send code" button to receive an SMS with the activation code. Then, follow the instructions according to the SMS and the SerpAPI website.





Obtaining the SerpAPI Key

5. Sign in and go to <https://serpapi.com/dashboard> , then copy the key at "Your Private API Key" section. Keep it secret. It will come in handy later.





Obtaining the NomicAPI Key

1. Go to the website <https://atlas.nomic.ai/> and click the "Get started with your Data" button. Then, create your account by setting up a username and password.





Obtaining the NomicAPI Key

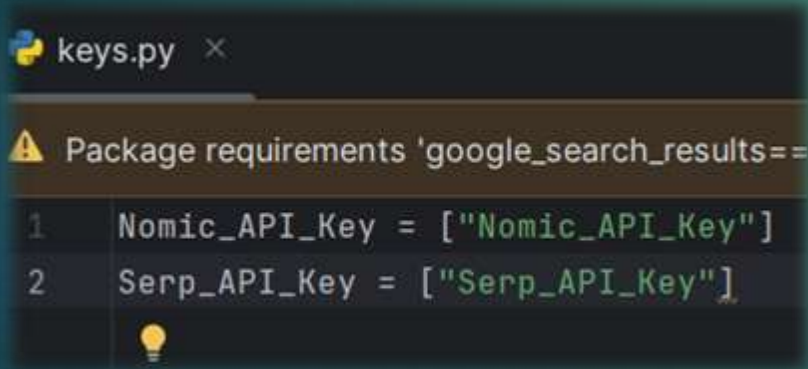
2. If you are logged in, go to the "API Keys" tab. Then, click the "Create new API Key" button. Keep it secret. It will come in handy later.





Initial Configuration

Please open the file `keys.py`. Paste the API keys: `Serp_API` and `Nomic_API` instead of `["Nomic_API_Key"]` and `["Serp_API_Key"]`

A screenshot of a code editor window titled 'keys.py'. It shows two lines of Python code: '1 Nomic_API_Key = ["Nomic_API_Key"]' and '2 Serp_API_Key = ["Serp_API_Key"]'. A yellow warning icon is visible on the left side of the code block.

```
keys.py ×  
⚠ Package requirements 'google_search_results==...  
1 Nomic_API_Key = ["Nomic_API_Key"]  
2 Serp_API_Key = ["Serp_API_Key"]  
💡
```

The API key should look like this:
`Nomic_API_Key = "my-nomic-api-key1234"`



Please open the file main.py. You will be prompted to enter a URL and a phrase

```
C:\Users\mateu\PycharmProjects\ISA\venv\Scripts\python.exe C:\Users\mateu\PycharmProjects\ISA\main.py
Enter your website URL: https://ratosnatura.pl/
Enter the keyword related to your website: miód naturalny
|
```

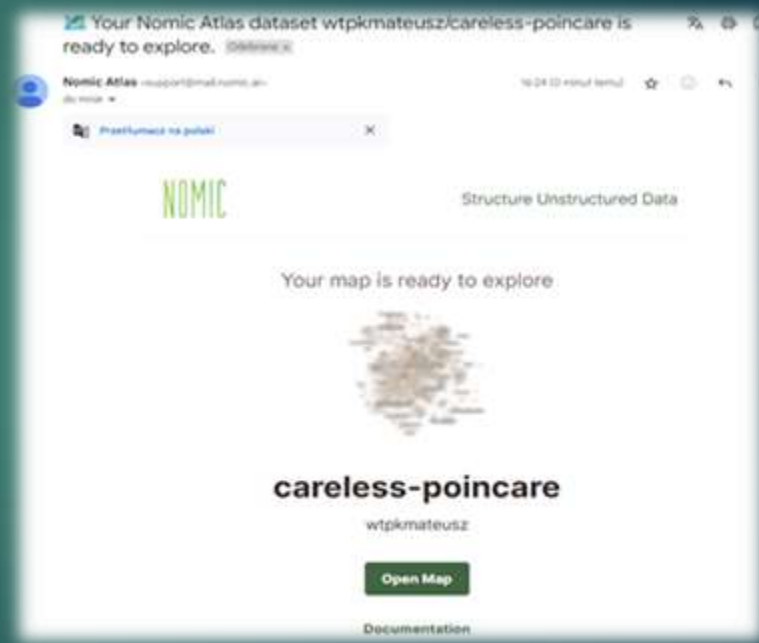


You will receive a notification that a new knowledge graph has been created in Nomic. Now, you need to wait up to 10 minutes for a confirmation email.

```
Run main x
11t [00:01, 1.07s/1t]
2024-10-13 16:20:39.573 | INFO | nomic.dataset:_add_data:1668 - Upload succeeded.
2024-10-13 16:20:39.573 | INFO | nomic.atlas:map_data:158 - 'wtpkmateusz/careless-poincare': Data upload succeeded to dataset
2024-10-13 16:20:42.829 | INFO | nomic.dataset:create_index:1262 - Created map 'careless-poincare' in dataset 'wtpkmateusz/careless-poincare'
careless-poincare: https://atlas.nomic.ai/data/wtpkmateusz/careless-poincare
Please wait for an email confirmation regarding the creation of the dataset (up to 10 minutes). You can check your email for the confirmation.
```

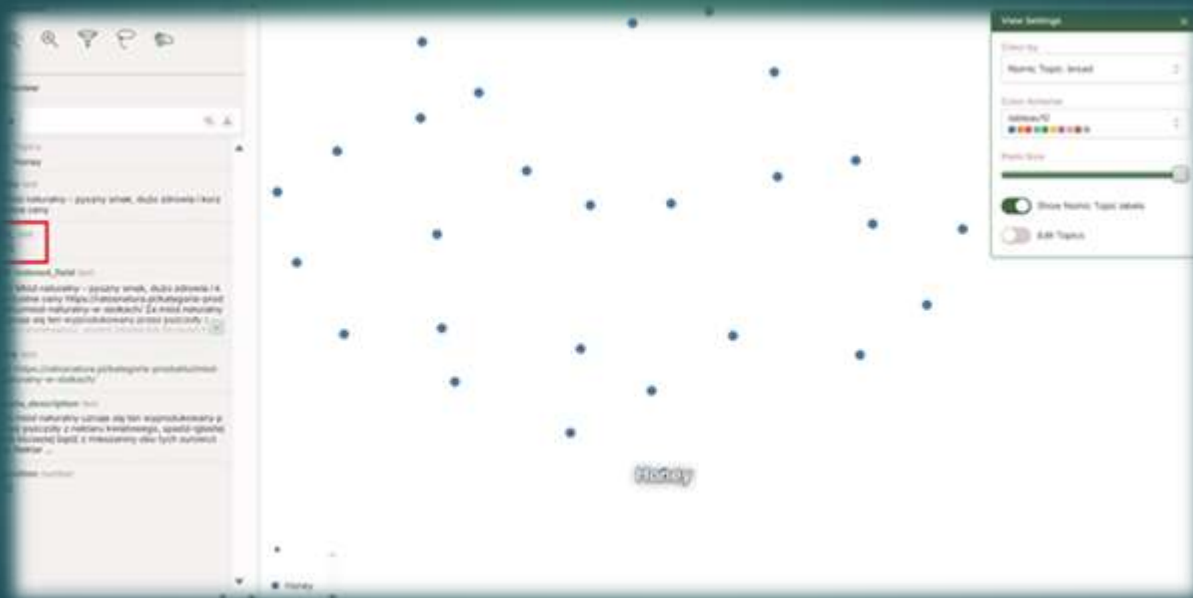


After processing, which will take up to 10 minutes, you will receive an email with the completed map from Nomic. Please open it using the email associated with your Nomic account. Copy the dataset ID name (in this case, it is careless-poincare).





Once the map is opened, copy your ID from our page (here, it is FQ).





Now you can perform various analyses by running the sec.py script.

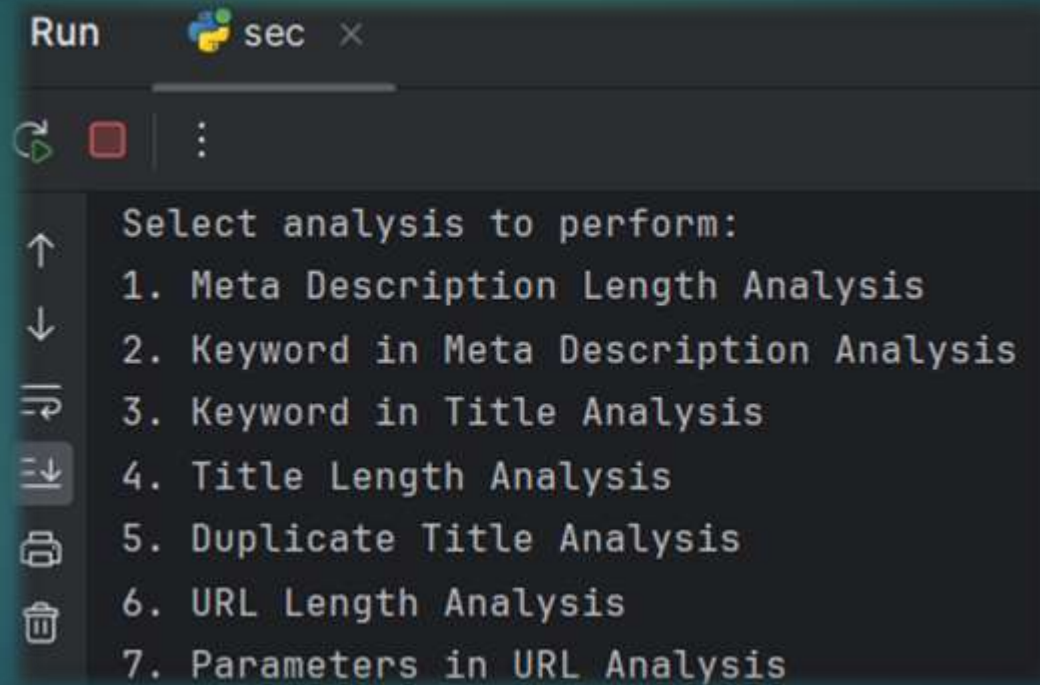
You need to provide the dataset ID from Nomic (here, it is careless-poincare) as well as the ID from our page (here, it is FQ).

```
Run  sec x
C:\Users\mateu\PycharmProjects\ISA\venv\Scripts\python.exe C:\Users\mateu\PycharmProjects\ISA\sec.py
Enter the dataset ID from Nomic: careless-poincare
2024-10-13 16:31:41.560 | INFO      | nomic.dataset.__init__:768 - Loading existing dataset 'wtpkmateusz/careless-poincare'.
careless-poincare: https://atlas.nomic.ai/data/wtpkmateusz/careless-poincare
Successfully loaded dataset with ID: careless-poincare
Loading analyses...
Enter the knowledge graph ID: FQ

Select analysis to perform:
```



Analyses in the project





Meta Description Length Analysis

The `meta_desc_length_analysis` function analyzes the lengths of meta descriptions in a DataFrame.

It first prompts the user for **the minimum and maximum lengths of the meta descriptions**, validating the input.

The function then calculates **the length of each description**, adding a new column to the DataFrame.

It filters the data to include only descriptions within the specified length range. The function prints **a message indicating the lengths of the meta descriptions within this range**.

It **highlights the relevant row** using the `highlight_rows_local` function.

A histogram is created, with the x-axis representing the lengths and the y-axis indicating whether at least one description exists for that length.

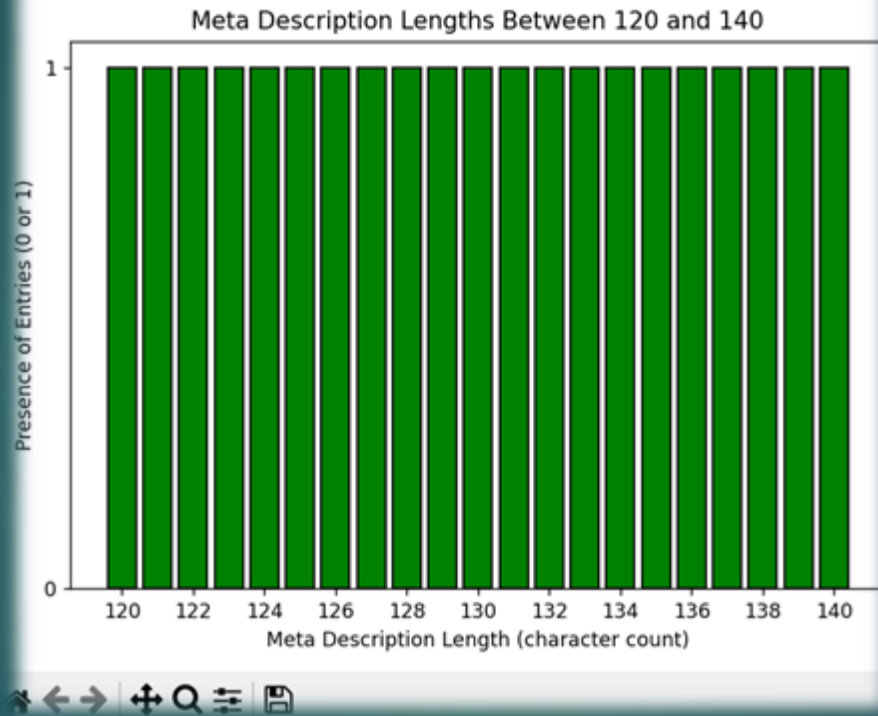
Finally, the user is prompted to press Enter to **return to the menu**.



```
Run  sec x
7. Parameters in URL Analysis
8. Enter new knowledge graph ID
9. Exit program
Enter your choice (1-9): 1
Enter the minimum meta description length: 120
Enter the maximum meta description length: 140
Meta descriptions with length between 120 and 140 characters:
['BQ', 135, 'Wycieraczki pod drzwi. Wycieraczka jako pierwsza wita cię w domu, ale zatrzymuje też błoto i piasek. W rezultacie podłoga i dywany w ...']
['Bg', 136, 'Wycieraczki Samochodowe Podłogowe na Allegro.pl - Zróżnicowany zbiór ofert, najlepsze ceny i promocje. Wejdź i znajdź to, czego szukasz!']
['Cg', 129, 'Naty te muszą być wypełnione roztworem dezynfekującym. Zmniejszają ryzyko rozprzestrzeniania się chorób bakteryjnych i wirusowych.']
['Dg', 134, 'Gumowa wycieraczka podłogowa Fingertip. Usuwa, zdrapuje i zatrzymuje bród. Podłoże antypodłigowe pomaga zachować pozycję wycieraczki.']
['GA', 139, 'Strona główna / Wycieraczka wejściowa. Filter. Filtry. Filtry. Pokaż [128] - Cancel - emco DIPLOMAT Premium z wkładką Care. Add to Compare.']
['CQ', 133, 'Wycieraczka bardzo odporna na warunki atmosferyczne. Świetnie usunie brud z obuwia zapobiegając rozprzestrzenianiu się zanieczyszczeń.']
```



Figure 1



Keyword in Meta Description Analysis



The `keyword_in_meta_description_analysis` function analyzes occurrences of a keyword in meta descriptions within a DataFrame.

The user is prompted to **enter a keyword to search for**, which is then escaped to avoid issues with regular expressions.

The function counts **occurrences of the keyword in the meta description column**, creating a new column with the count.

It then prints **information about the number of occurrences of the keyword**.

Rows containing the keyword are highlighted using the `highlight_rows_local` function.

The function separates counts for cases with 0 occurrences and 1 or more occurrences.

A bar plot is created, with the x-axis representing the number of occurrences and the y-axis showing the number of entries. Finally, the user is prompted to press Enter to return to the menu.



Run

sec x



9. Exit program



Enter your choice (1-9): 2



Enter the keyword to search in the meta description: *wycieraczka podłogowa*



Number of occurrences of the keyword 'wycieraczka podłogowa' in the meta description:



['HQ', 0, 'Aximo.pl to dystrybutor przemysłowych wykładzin obiektowych i wycieraczek podłogowych']



['FA', 0, 'Wycieraczki przednie Mata wejściowa Wewnątrz Na zewnątrz Absorpcja wody Maty podłogowe']

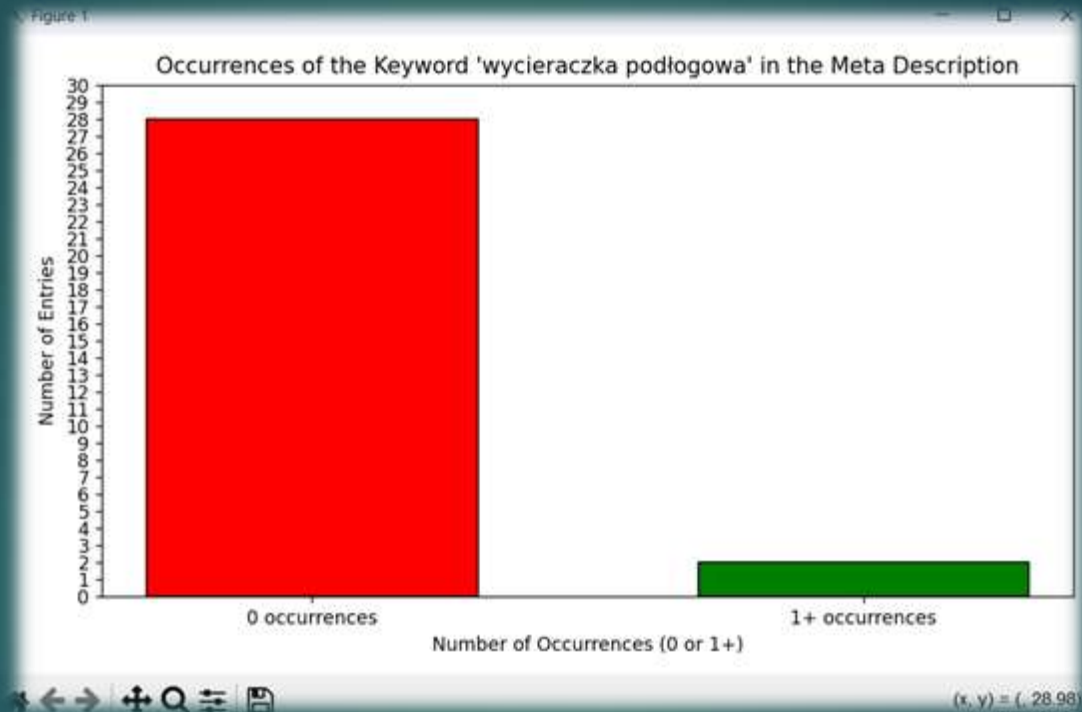
['Gw', 0, 'Płytki podłogowe. Wzór płytki. Imitujące drewno ... Wycieraczki do przedpokojuWnętrze']

['GQ', 0, 'Wycieraczki pod drzwi. Wycieraczka jako pierwsza wita cię w domu, ale zatrzymuje brzo']

['Gg', 0, 'Wytrzymała, skutecznie czyszcząca każdą podeszwę wycieraczka dywanikowa półokrągła']

['FQ', 0, '... podłogowe - dywany i wycieraczki - sklep online ... Wycieraczka Wejściowa Przedpokoju']

['Bw', 0, 'Chłonna wycieraczka wejściowa klasyczna czy nowoczesna, zewnętrzna lub wewnętrzna']





Keyword in Title Analysis

The `keyword_in_title_analysis` function analyzes **occurrences of a keyword in titles** within a DataFrame.

The user is prompted to **enter a keyword to search for**, which is then escaped to avoid issues with regular expressions.

The function counts **occurrences of the keyword in the title column**, creating a new column with the count.

It then prints **information about the number of occurrences of the keyword in the titles**.

Rows containing the keyword are highlighted using the `highlight_rows_local` function.

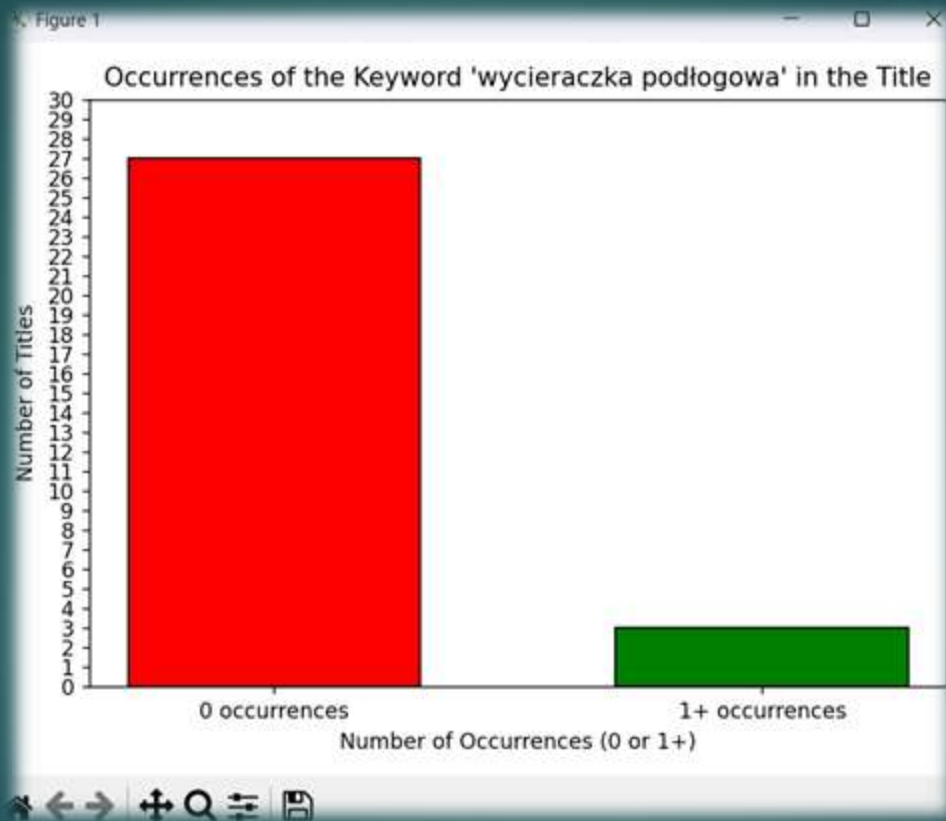
The function separates counts for cases with 0 occurrences and 1 or more occurrences.

A bar plot is created, with the x-axis representing the number of occurrences and the y-axis showing the number of titles.

Finally, the user is prompted to press Enter to return to the menu.



```
Run  sec x
Enter your choice (1-9): 3
Enter the keyword to search in the title: wycieraczka podłogowa
Number of occurrences of the keyword 'wycieraczka podłogowa' in the title:
['HQ', 0, 'Komfortowe wykładziny dywanowe i wycieraczki dla firm - Aximo.pl']
['FA', 0, 'Wycieraczka Pod Drzwi Wejściowe']
['Gw', 0, 'Wycieraczki']
```





Title Length Analysis

The `title_length_analysis` function analyzes title lengths within a DataFrame, allowing the user to define a range of lengths.

The user is prompted to enter **minimum and maximum title lengths**, and the function provides error handling to ensure valid input.

It then calculates **the lengths of the titles** and filters the DataFrame based on the specified range. Titles that fall within the defined length are displayed.

Rows matching the criteria are highlighted using the `highlight_rows_local` function.

The function counts occurrences of each title length in the filtered DataFrame and creates **a histogram**, where the bar colors indicate the presence or absence of occurrences. The x-axis represents title lengths, and the y-axis shows the number of titles.

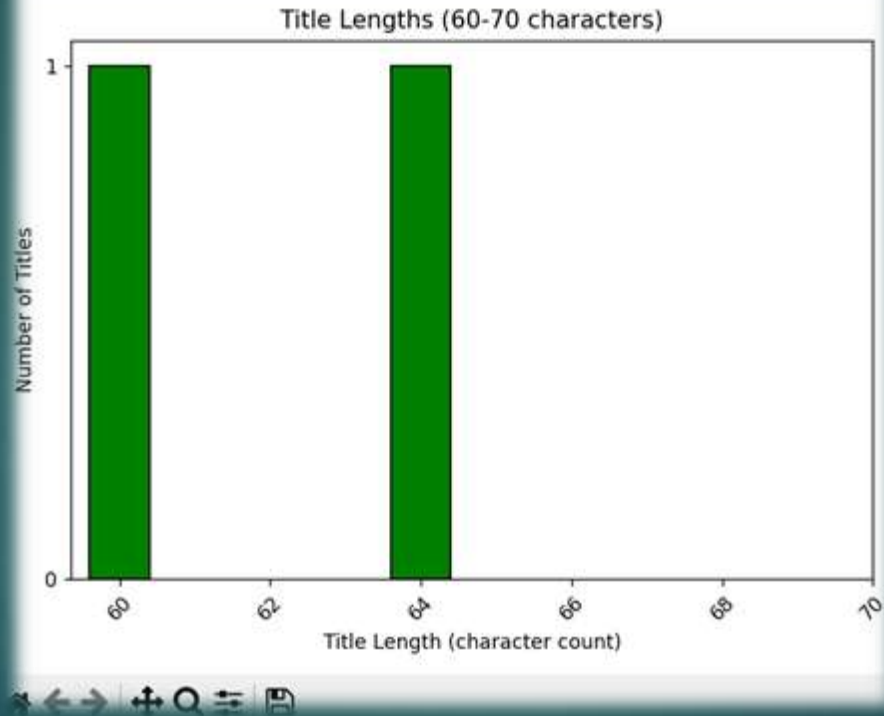
Finally, the user is prompted to press Enter to return to the menu.



```
Run  sec x
:
5. Duplicate Title Analysis
6. URL Length Analysis
7. Parameters in URL Analysis
8. Enter new knowledge graph ID
9. Exit program
Enter your choice (1-9): 4
Enter the minimum title length: wycieraczka podłogowa
Invalid input. Please enter numeric values for lengths.
Enter the minimum title length: 60
Enter the maximum title length: 70
Titles with length between 60 and 70 characters:
['HQ', 64, 'Komfortowe wykładziny dywanowe i wycieraczki dla firm - Aximo.pl']
['Fw', 60, 'Wycieraczka mata wejściowa Perła różne kolory i rozmiary ...']
```



Figure 1





Duplicate Title Analysis

The `duplicate_title_analysis` function identifies titles that are duplicated within the DataFrame.

It **searches for duplicate titles and displays them** along with their ID, link, and meta description information using the `highlight_rows_local` function.

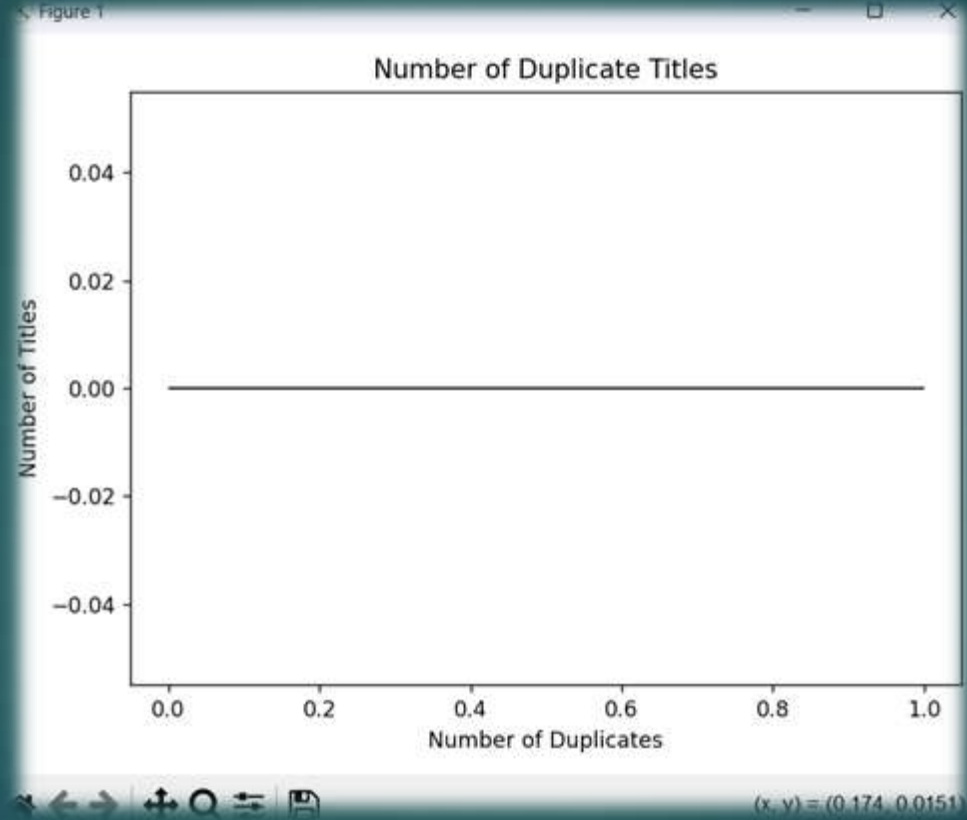
The function then counts **occurrences of each title** to identify those that appear more than once.

It generates a **histogram** showing the number of titles based on the number of their duplicates. The x-axis represents the number of duplicates, while the y-axis shows the number of titles. The histogram is colored red with a black edge for better visualization.

After displaying the histogram, the user is prompted to press Enter to **return to the menu**. This function provides an easy way to identify and analyze problematic titles in the dataset.



```
Run  sec x
Select analysis to perform:
1. Meta Description Length Analysis
2. Keyword in Meta Description Analysis
3. Keyword in Title Analysis
4. Title Length Analysis
5. Duplicate Title Analysis
6. URL Length Analysis
7. Parameters in URL Analysis
8. Enter new knowledge graph ID
9. Exit program
Enter your choice (1-9): 5
Duplicate Titles:
```





URL Length Analysis

The `url_length_analysis` function analyzes the lengths of URLs in a DataFrame. It first prompts the user to input **the minimum and maximum URL lengths**, providing error handling for invalid input.

The function then **calculates the length of each URL** by adding a new column to the DataFrame.

It filters the data to include only those URLs whose lengths fall within the specified range.

The function displays the filtered URLs along with their lengths using the `highlight_rows_local` function.

It creates **a histogram** of URL lengths, with the x-axis representing the lengths and the y-axis showing the number of URLs. The x-axis ticks are set to increment by 5, while the y-axis shows whole numbers.

After completion, the user is prompted to press Enter to **return to the menu**.

 sec 

7. Parameters in URL Analysis

8. Enter new knowledge graph ID

9. Exit program

```
Enter your choice (1-9): 6
```

```
Enter the minimum URL length: 120
```

```
Enter the maximum URL length: 140
```

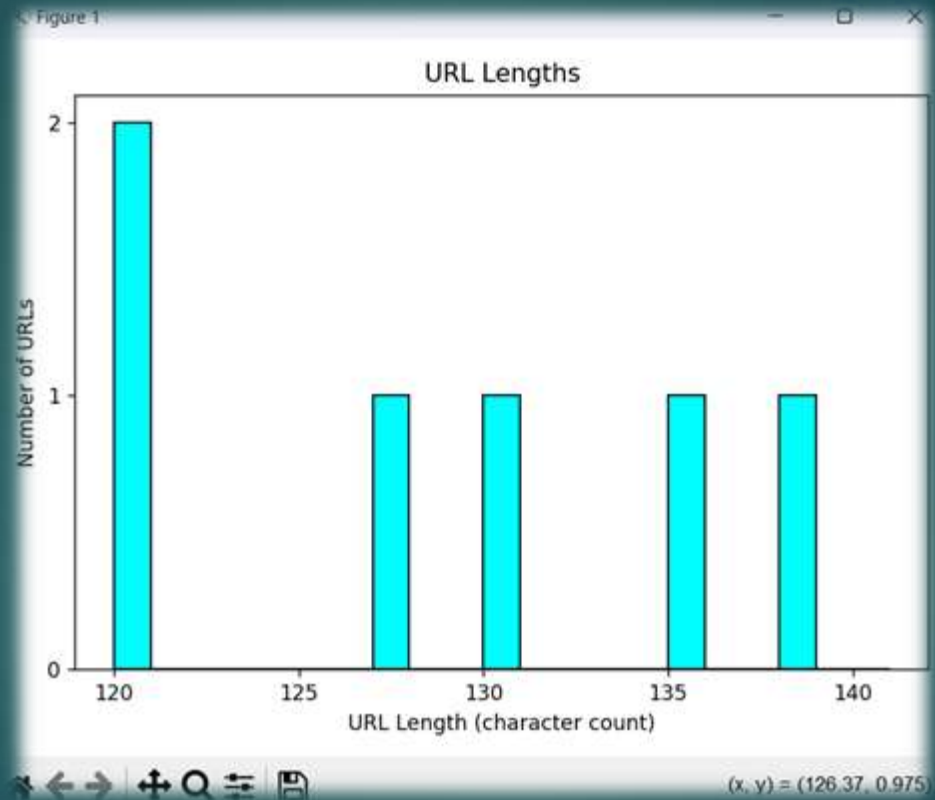
Filtered URL Lengths:

```
['AA', 130, 'https://allegro.pl/kategoria/dywany-i-dywaniki-wycierac
```

```
[ 'AQ', 120, 'https://esilver.com.pl/wycieraczki-wejscowe-c-175.htm'
```

```
['GA', 135, 'https://www.empik.com/wycieraczka-do-butow-wejsciowa-p
```

```
['CA', 127, 'https://matyobiektove.pl/produkt/mata-podlogowa-wyciera
```





Parameters in URL Analysis

The `params_in_url_analysis` function analyzes URLs in a DataFrame for parameters. It begins by **checking if a URL contains parameters**, adding a new column to indicate this.

Next, it counts the number of parameters in each URL by splitting at the question mark (?).

The function extracts the query parameters for **display and presents the URLs with parameters, showing their IDs and values**. It uses the `highlight_rows_local` function to highlight the relevant rows in the DataFrame, displaying the ID, URL, and parameter count.

The function then creates **a histogram** of the number of parameters in the URLs, with the x-axis representing the number of parameters and the y-axis showing the number of URLs. Both axes are configured to show whole numbers.

After completion, the user is prompted to press Enter to **return to the menu**.



Run

sec

Enter your choice (1-9): 7

URLs with Parameters:

ID: FQ, URL: <https://sklepoxide.pl/wycieraczki?srsltid=AfmB0or8zv5Yk0r-ey49EZV>, Query Parameter: srsltid=AfmB0or8zv5Yk0r-ey49EZV

ID: Bw, URL: <https://sklepoxide.pl/pl/c/Wycieraczki-gumowe/138?srsltid=AfmB0ooG-97awADext0YUBA1Y0-RwYmBaAznGbEhM7XUkhtxs5WRtaq>, Query Parameter: srsltid=AfmB0ooG-97awADext0YUBA1Y0-RwYmBaAznGbEhM7XUkhtxs5WRtaq

ID: AA, URL: <https://allegro.pl/kategorie/dywany-i-dywaniki-wycieraczki-110184?srsltid=AfmB0ooG-YVUSLlYThcmeas0TJNT86fnGftz0ssR1Sa03ZY-qphJH0e5>, Query Parameter: srsltid=AfmB0ooG-YVUSLlYThcmeas0TJNT86fnGftz0ssR1Sa03ZY-qphJH0e5

ID: Aw, URL: <https://sklepurska.pl/wycieraczki?srsltid=AfmB0ooGwYIvvbAnehn12C3czMc07TJ5b54XZFw6un-K2mzx8cPvcI>, Query Parameter: srsltid=AfmB0ooGwYIvvbAnehn12C3czMc07TJ5b54XZFw6un-K2mzx8cPvcI

ID: AQ, URL: <https://esilver.com.pl/wycieraczki-welosciowe-c-175.html?srsltid=AfmB0ooGCPaksn0N4dF1AYJcda30BYB15N7YJmW52hXaya-AGZ5mGTiH>, Query Parameter: srsltid=AfmB0ooGCPaksn0N4dF1AYJcda30BYB15N7YJmW52hXaya-AGZ5mGTiH

ID: Fg, URL: <https://ergosport.pl/beq-produkty/wycieraczki-na-wyniar/?srsltid=AfmB0ooV2Ugru0wHMoUeehG7P7RfYeuqJyI-whwv22u7bk10esfzQV0I>, Query Parameter: srsltid=AfmB0ooV2Ugru0wHMoUeehG7P7RfYeuqJyI-whwv22u7bk10esfzQV0I

ID: Bg, URL: <https://allegro.pl/listing?string=wycieraczki%20sianochodowe%20pod%25k87ogowe&srsltid=AfmB0oo8b1155FP2L08KHQAYalrUjPvY5nPicKHQn6kFY133XxQsp8h>, Query Parameter: srsltid=AfmB0oo8b1155FP2L08KHQAYalrUjPvY5nPicKHQn6kFY133XxQsp8h

ID: BA, URL: <https://matypodlogowe.eu/maty-asieciowe/?srsltid=AfmB0oo01fXo-tX6B--Y0B-Re535PB-ps7CS10UwedvvyF-h60IWYUn>, Query Parameter: srsltid=AfmB0oo01fXo-tX6B--Y0B-Re535PB-ps7CS10UwedvvyF-h60IWYUn

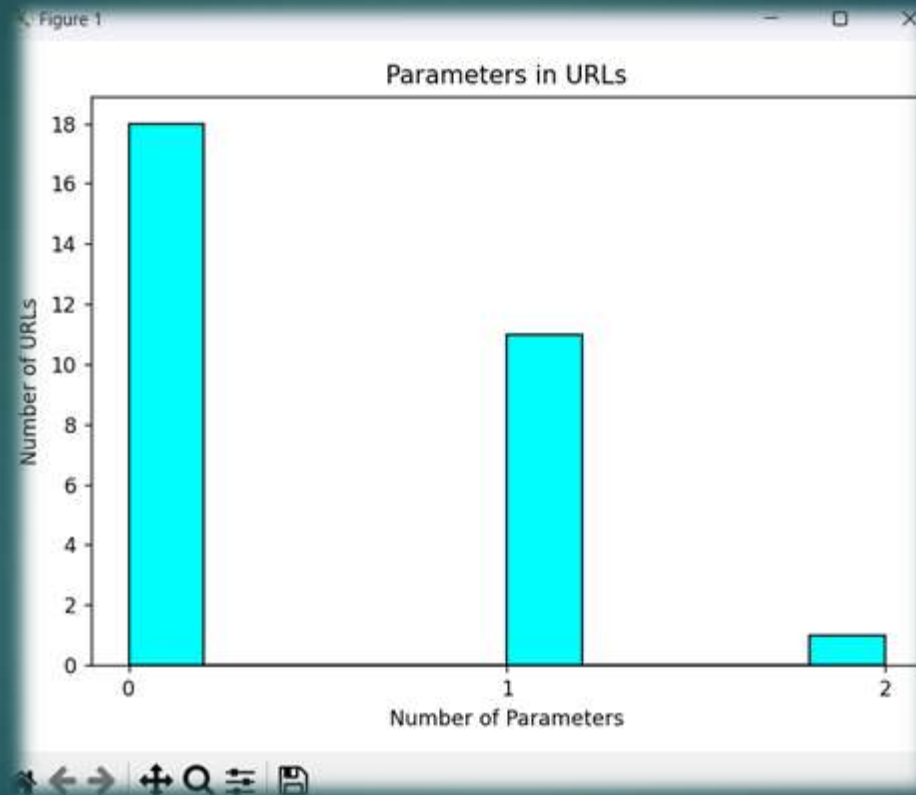
ID: CA, URL: <https://matyobiektove.pl/produkty/maty-podlogowe-wycieraczka-2/?srsltid=AfmB0oo2In4e9IhoR30o-qIaNMt0LR8tgeK7Ihn7I8xCKhPoxiQInVE>, Query Parameter: srsltid=AfmB0oo2In4e9IhoR30o-qIaNMt0LR8tgeK7Ihn7I8xCKhPoxiQInVE

ID: Ag, URL: https://stannats.pl/wycieraczki-welosciowe-tekstylne?srsltid=AfmB0oo8T9CEsRrqsV0qCP7eI-7x8P3DYs_uHwZnu0V1Xee3vzIv0P5, Query Parameter: srsltid=AfmB0oo8T9CEsRrqsV0qCP7eI-7x8P3DYs_uHwZnu0V1Xee3vzIv0P5

ID: Cg, URL: <https://www.happymendel.pl/podlogowe-wycieraczka-mata-kleista-wielowierszowa?srsltid=AfmB0oo4M0ZQC0xfalI-9t9IR50UfSu14Iyu3d1C3I0rAPcHUSJout-y>, Query Parameter: srsltid=AfmB0oo4M0ZQC0xfalI-9t9IR50UfSu14Iyu3d1C3I0rAPcHUSJout-y

ID: Dg, URL: <https://sklep.greenerservice.pl/pl/wycieraczki-i-maty-asieciowe/841-gumowe-wycieraczka-podlogowe-fingertip.html?srsltid=AfmB0ooJCCEAvayJp15Iva6CXWf-i>, Query Parameter: srsltid=AfmB0ooJCCEAvayJp15Iva6CXWf-i

["nq", "https://www.xinio.pl/'. n]





Project summary

- ❖ Many meta descriptions are either too short or too long, which can impact visibility in search engines. The ideal meta description length is typically between 150 and 160 characters, and analyses can indicate which entries require optimization.
- ❖ Analyses reveal which keywords are most frequently used in meta descriptions and titles. A high occurrence of a specific keyword may indicate the popularity of a topic, but it may also suggest the need for further content optimization to improve SEO rankings.
- ❖ Many entries in the database have duplicate titles, which can affect SEO and user experience. Identifying duplicate titles can help in editing or removing them, thereby improving the uniqueness of the content.
- ❖ Analyzing URL length may reveal that many links are excessively long. Long URLs can be less user-friendly and inefficient for SEO. Optimizing URL length should be considered to make them more concise and easier to remember.
- ❖ The number of parameters in URLs can indicate a complex site structure. Too many parameters can negatively impact indexing by search engines. Identifying such URLs can assist in simplifying the structure and improving SEO performance.
- ❖ Data analysis may uncover certain patterns or trends in user behavior, such as preferences for specific keywords or titles. This information can be valuable for future marketing strategies and in creating content that better meets audience needs.



Recommendations for business

- ❖ Revise meta descriptions to ensure they fall within the optimal length (150-160 characters). This can improve click-through rates from search engine results and enhance visibility.
- ❖ Conduct keyword research to identify relevant and high-traffic keywords. Integrate these keywords more effectively into titles and meta descriptions, ensuring they align with user intent.
- ❖ Review the database for duplicate titles and take action to make them unique. This can help improve SEO rankings and avoid confusion for users.
- ❖ Simplify lengthy URLs by removing unnecessary parameters and making them more concise. Aim for user-friendly URLs that are easy to read and remember, which can positively impact SEO.
- ❖ Analyze URLs for excessive parameters and consider restructuring the website to minimize them. This can improve site indexing by search engines and enhance user experience.
- ❖ Use the insights gained from analyzing user behavior patterns and trends to inform content strategy. Create targeted content that aligns with audience interests and search behavior to drive engagement and conversion.